# A New Auxiliary Model Approach For Fermionic Criticality

## Insights on Mottness in Strongly Correlated Systems

**ABHIRUP MUKHERJEE**

*Presubmission Open Seminar*

IISER KOLKATA

**Emergent Phenomena in Quantum Matter** group
IISER Kolkata

EPQM

# Acknowledgements

# Publications

**New J. Phys. 25 113011**. **AM**, N S Vidhyadhiraja, A Taraphder, S Lal. (2023)

**J. Phys. A: Math. Theor. 57 275401**. **AM**, S Patra, S Lal. (2024)

**New J. Phys. 28 013503**. **AM**, S R. Hassan, A Mukherjee, N S. Vidhyadhiraja, A Taraphder, S Lal. (2026)

**in progress**. **AM**, S Lal. (2026)

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**Phys. Rev. B 105, 085119**. A Mukherjee, **AM**, N S. Vidhyadhiraja, A Taraphder, S Lal. (2022)

**J. Phys.: Condens. Matter 35 315601**. S Patra, **AM**, A Mukherjee, N S Vidhyadhiraja, A Taraphder, S Lal. (2023)

**Phys. Rev. B 111, L140414**. A Kumar N M, S Mukherjee, **AM**, A Punjal, S Purwar, T Setti, S Prabhu S., S Lal, N Kamaraju. (2025)

**arXiv:2602.20600**. D Debata, **AM**, S Lal. (2026)

# Outline of Thesis

**Chap 1**. Introduction

**Chap 2**. Methods and Preliminaries

**Chap 3**. Kondo frustration via charge fluctuations: Mott localisation in $d = \infty$ [New J. Phys. 25 113011]

**Chap 4**. A New Auxiliary Model Mapping For Correlated Lattice Models [New J. Phys. 28 013503]

**Chap 5**. Mott Criticality as the Confinement Transition of a Pseudogap-Mott Metal in $d = 2$ [New J. Phys. 28 013503]

**Chap 6**. Competing tendencies In Heavy Fermion Systems: Kondo screening vs. Mott localisation [in progress]

**Chap 7**. Holographic entanglement renormalisation for fermionic quantum matter [J. Phys. A: Math. Theor. 57 275401]

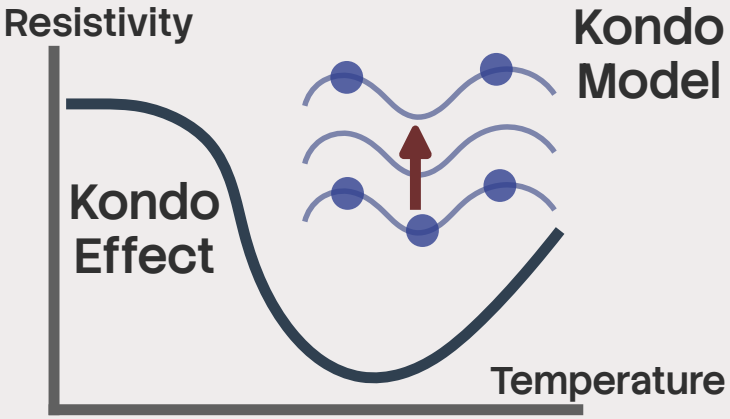
**Chap 8**. Conclusions

# Introduction To Fermionic Criticality and Quantum Materials

# What Do Condensed Matter Theorists Do?

## Explain Experiments

- Build and solve theoretical models for experiments
- Compute measurables to explain observations
- Kondo effect, quantum Hall effect, superconductivity



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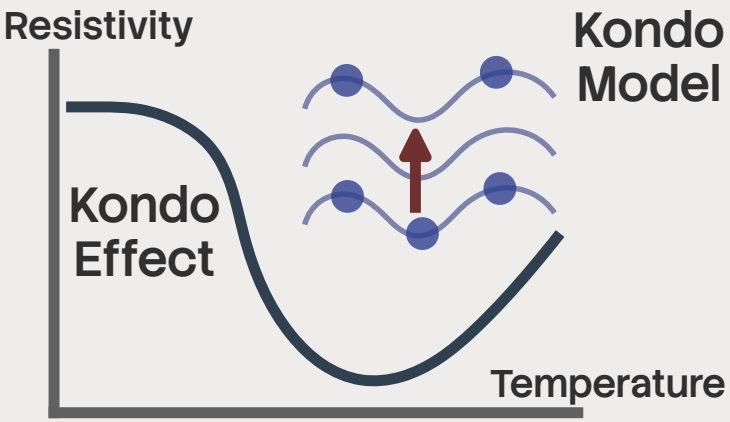
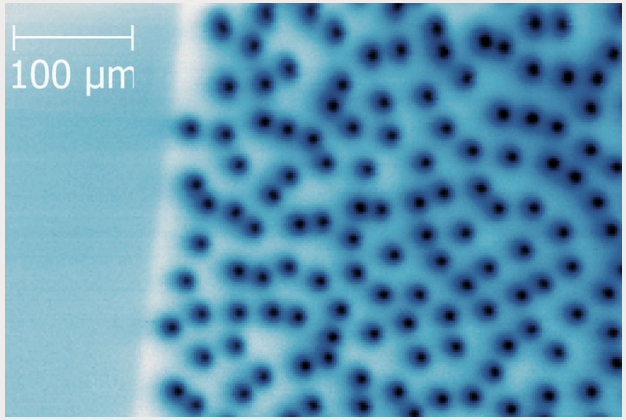



## Predict Experiments

- Identify interesting theoretical models for new systems
- Predict novel phenomena by tweaking models and parameters
- Abrikosov vortex lattice, spin liquids



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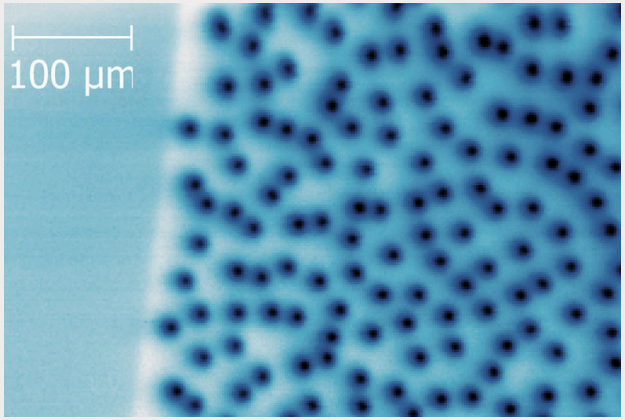


## Classify experiments

- organize experimental signatures into universal principles
- come up with frameworks that explain a large class of phenomena
- Examples: Landau theory of phase transition

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Class of materials that **cannot** be well-approximated by “classical” description

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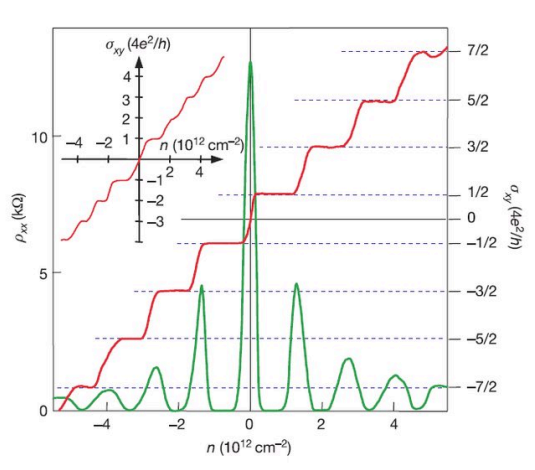


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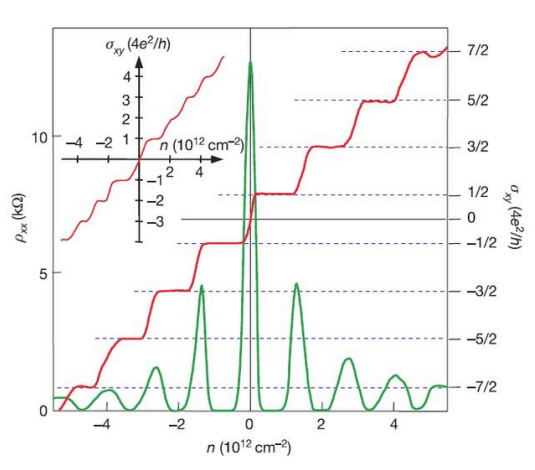


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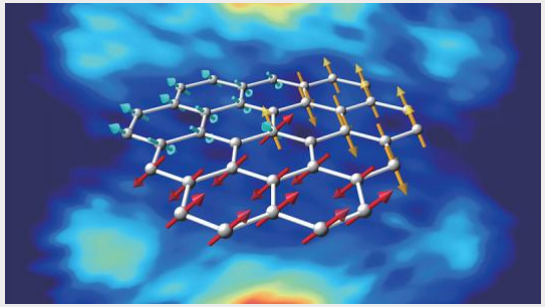

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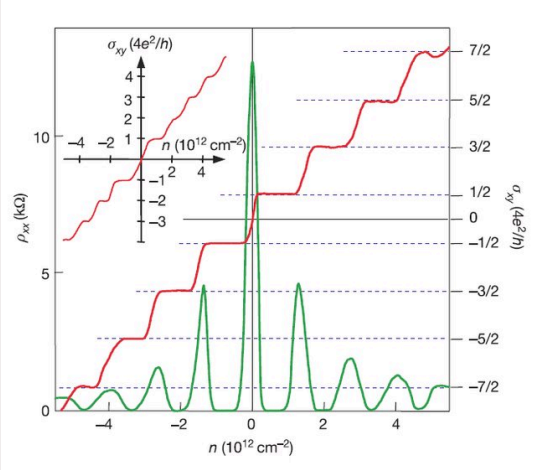


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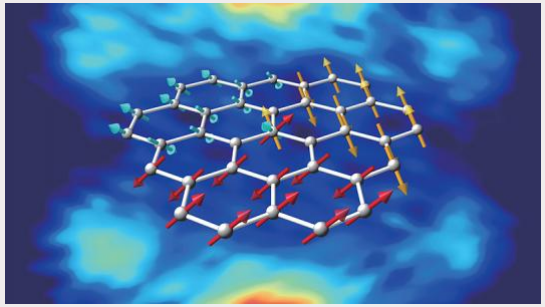

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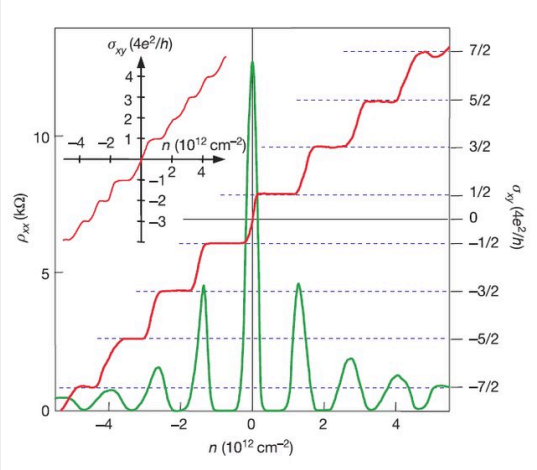


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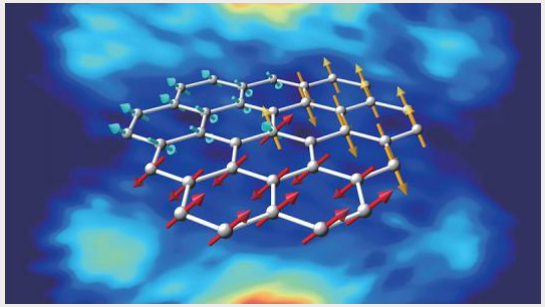

Needs dramatically **new ideas** and methods!

- **non-local/collective** order parameters for transition
- ability to work with a large number of **interacting particles**
- ability to work with **emergent** degrees of freedom

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Materials with strong **local** electronic correlation, **low dimensionality**, multiple electronic **orbitals**

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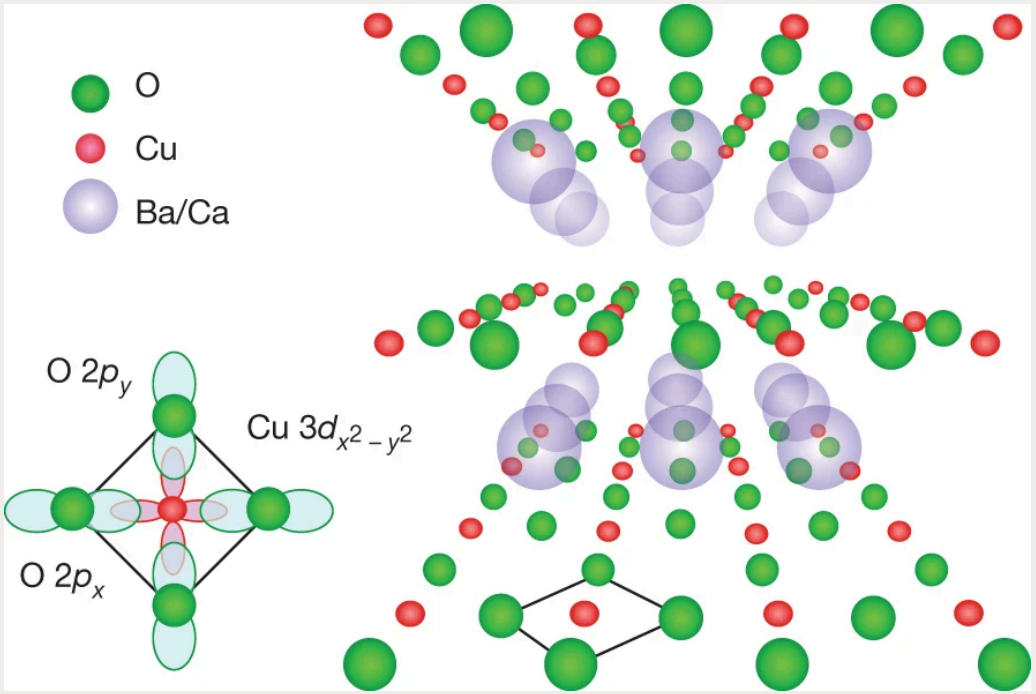

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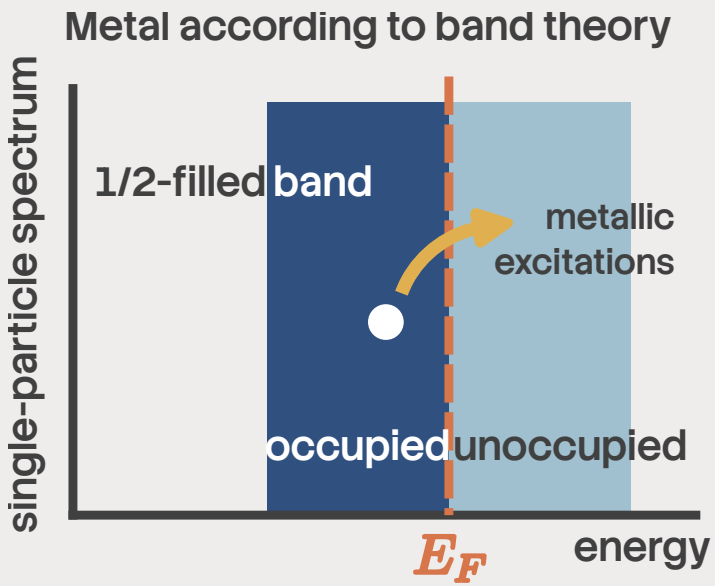

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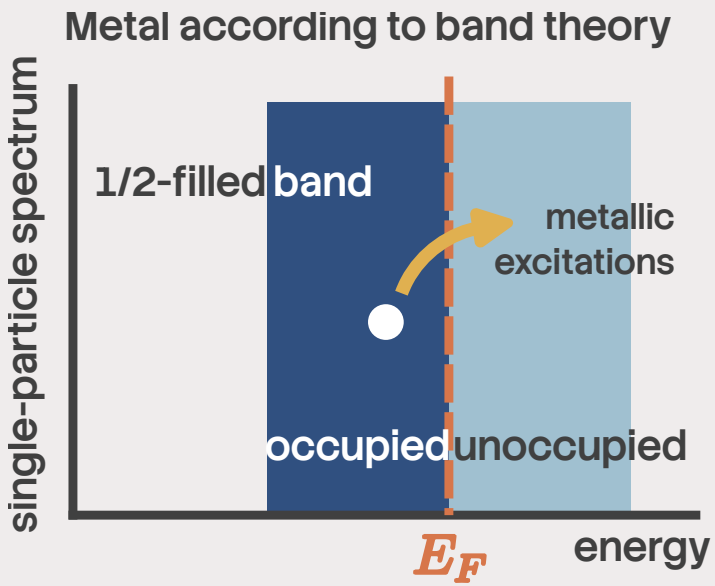


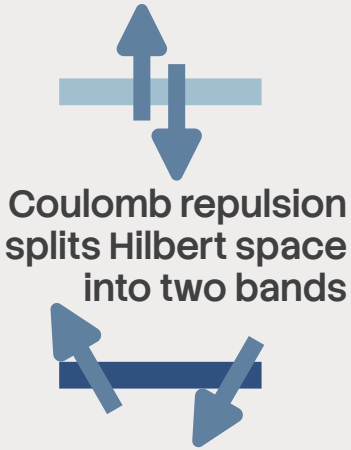

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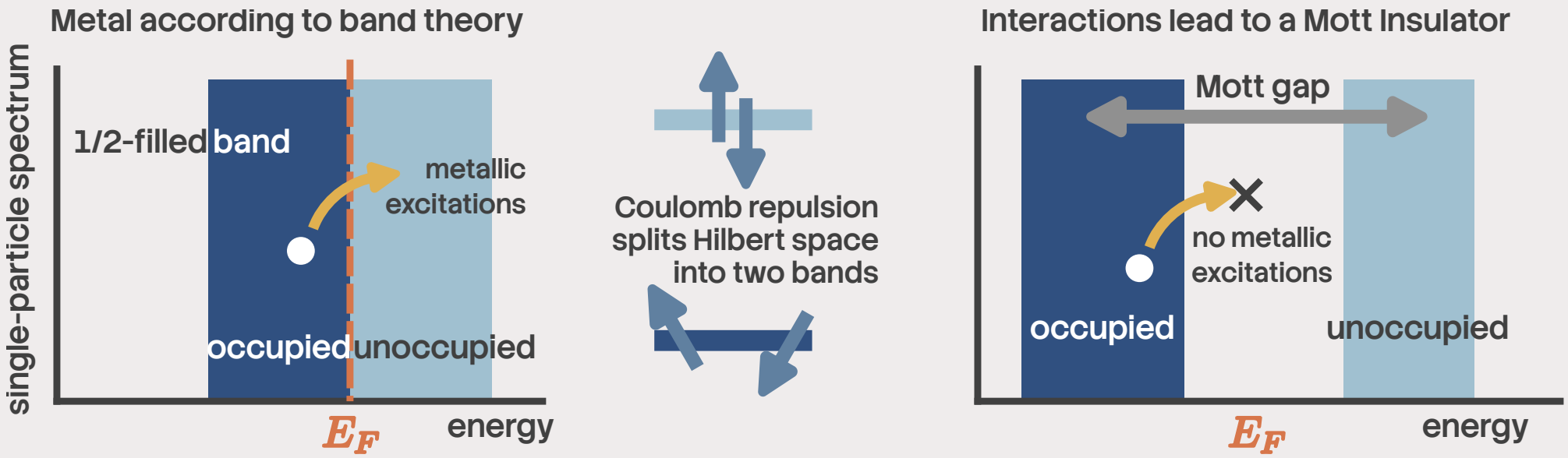

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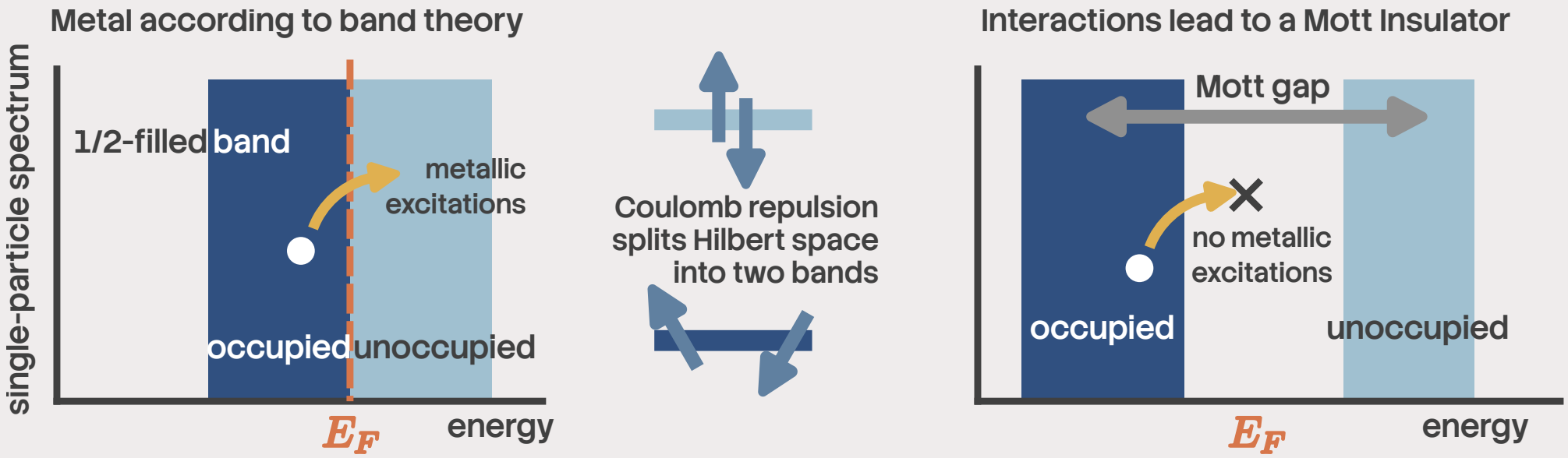


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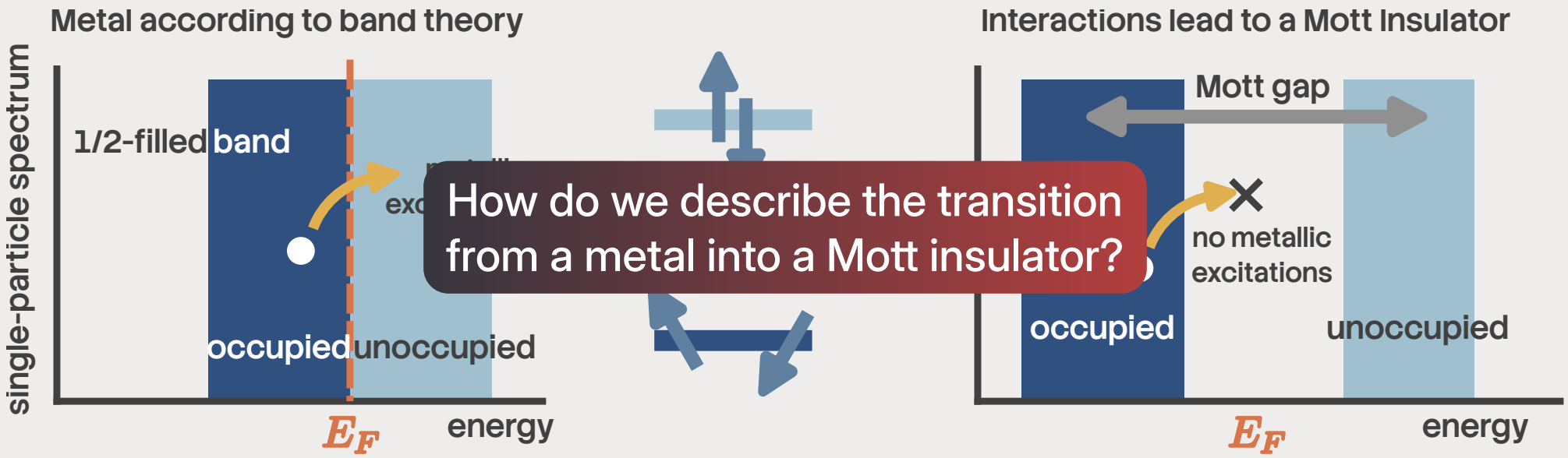


How do we describe the transition from a metal into a Mott insulator?

- Not accessible from 1-particle theory
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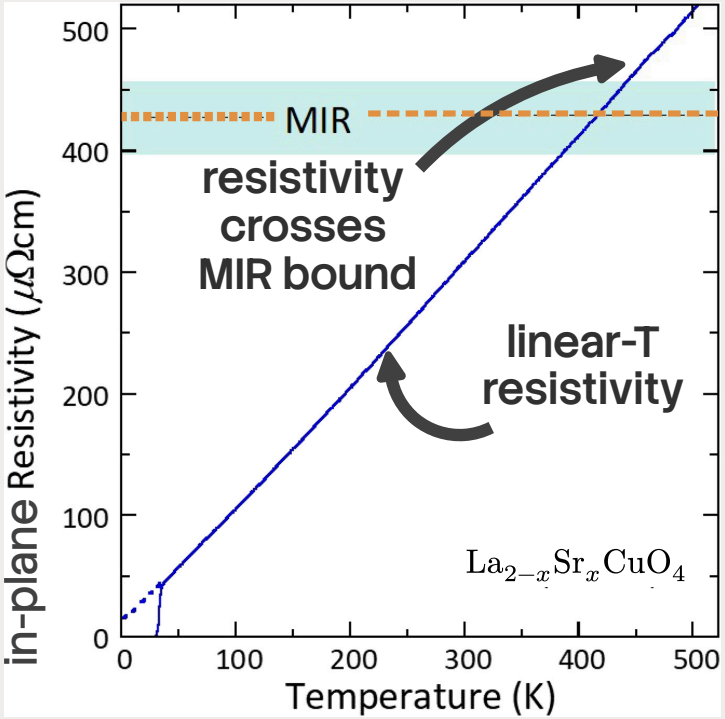

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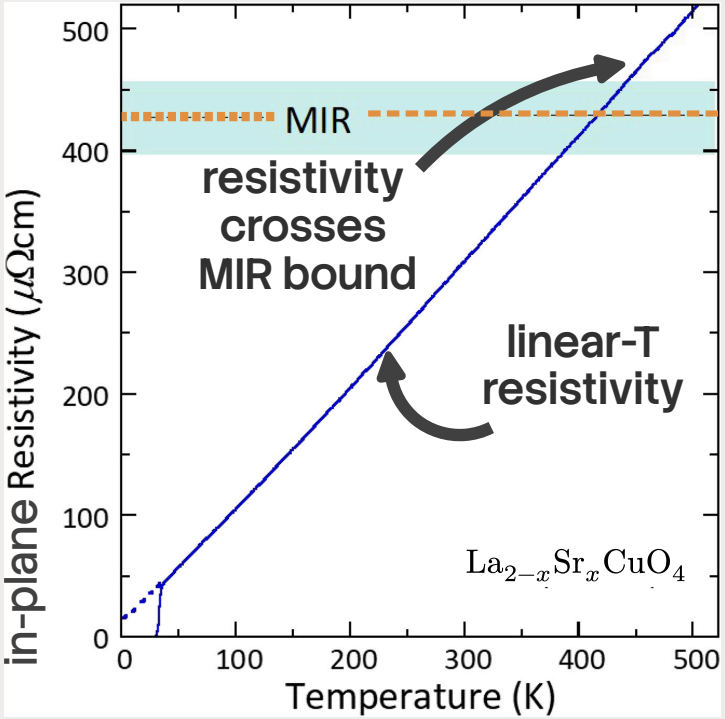


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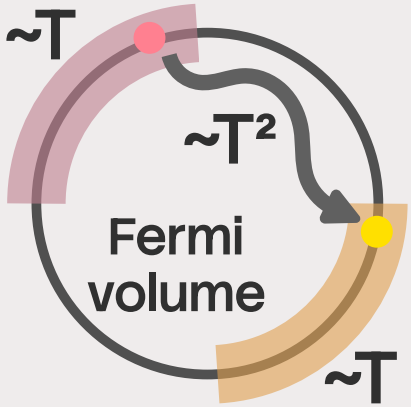


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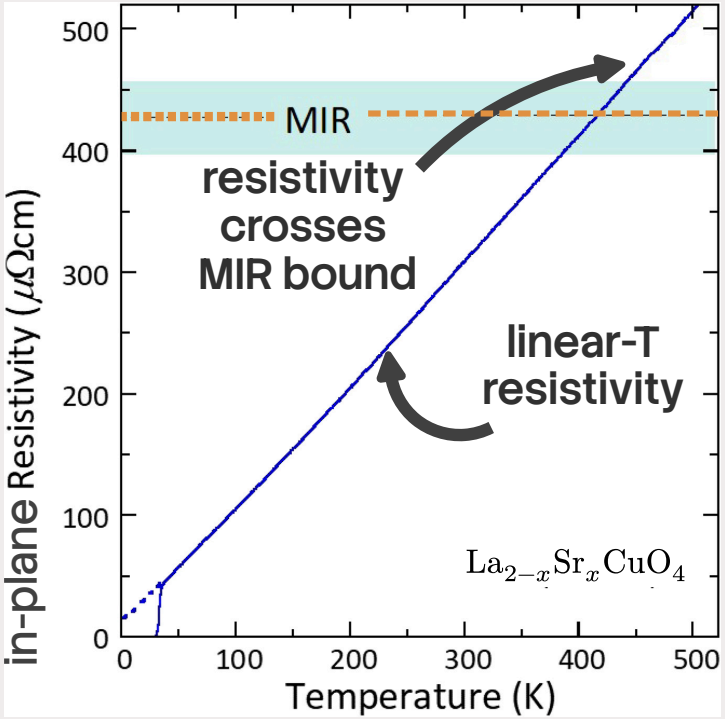


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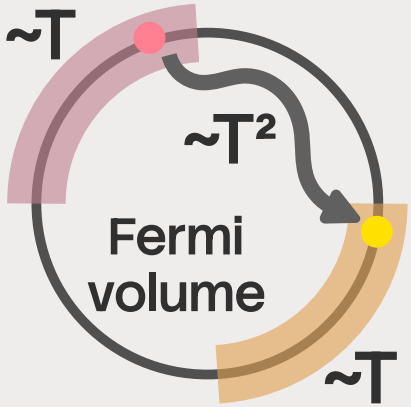


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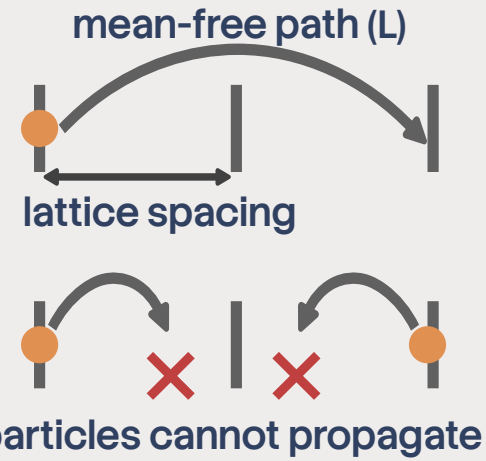
Single particles must propagate finite number of lattice spacing



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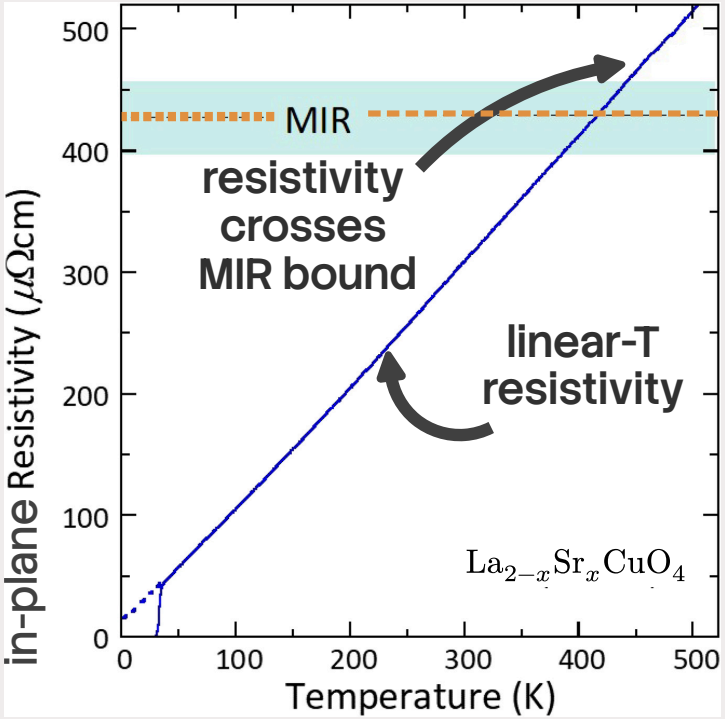


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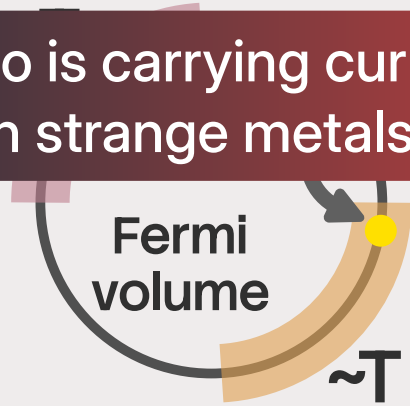
**Who is carrying current in strange metals?**

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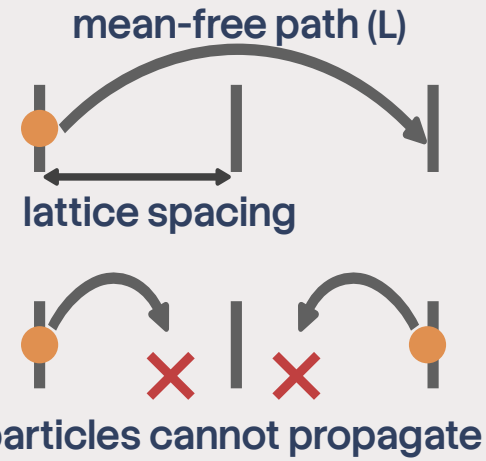

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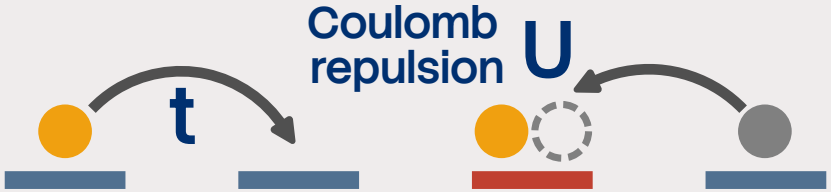

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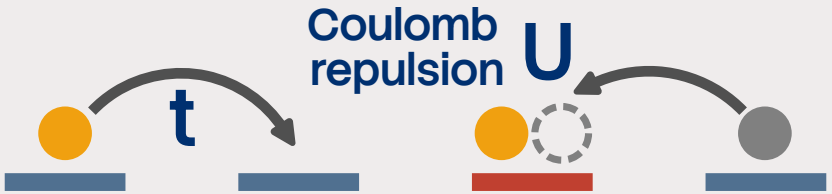


**Hamiltonian:** $-t\sum_{i,j,\sigma}(c^{\dagger}_{i,\sigma}c_{j,\sigma} + \text{h.c.}) + \sum_i U n_{i,\uparrow}n_{i,\downarrow}$

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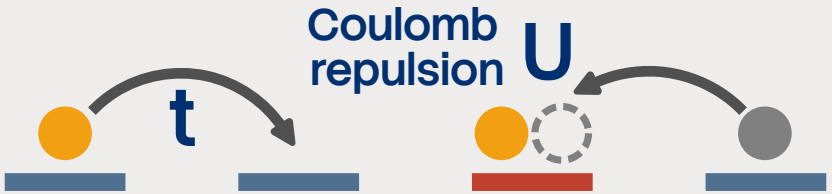


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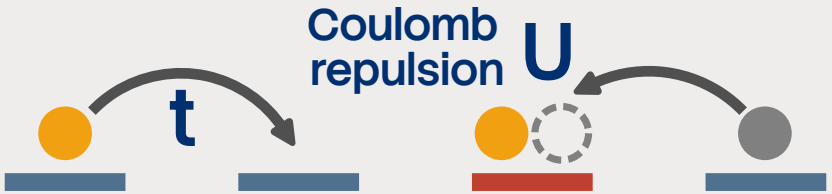


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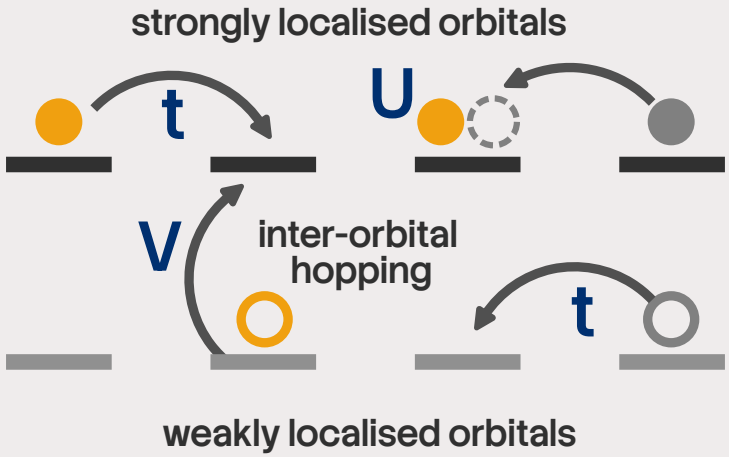


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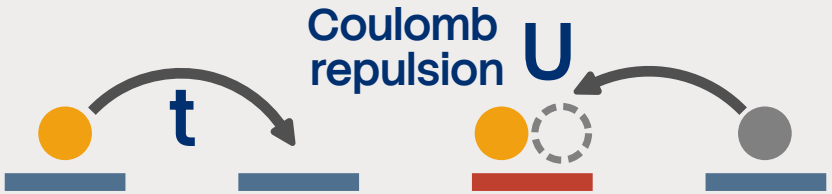


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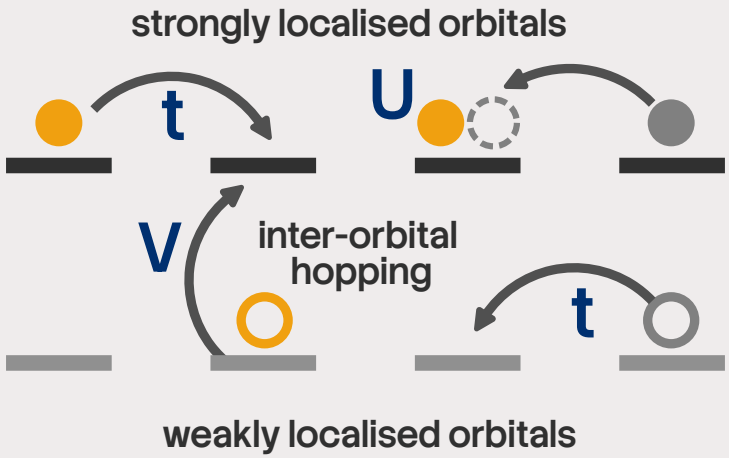


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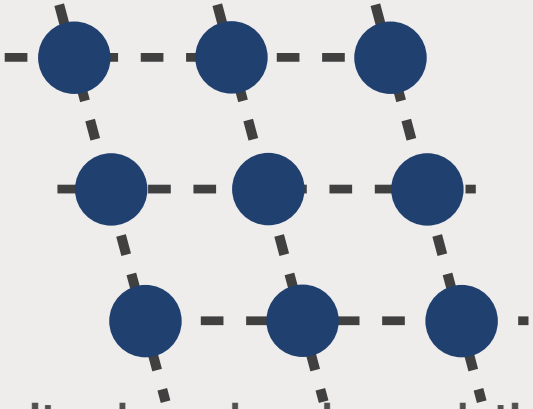
**Hamiltonian:** $H_{\text{Hub}} + H_{\text{hop}} + V\sum_{i,\sigma} f^{\dagger}_{i,\sigma}c_{i,\sigma} + \text{h.c.}$

# Quantum Impurity Models as Auxiliary Models

Lattice models of correlated electrons are **hard to solve** directly.

Correlated lattice model



All sites have local correlation

**Idea**: replace full problem with a simpler **auxiliary model** that captures the local correlations.

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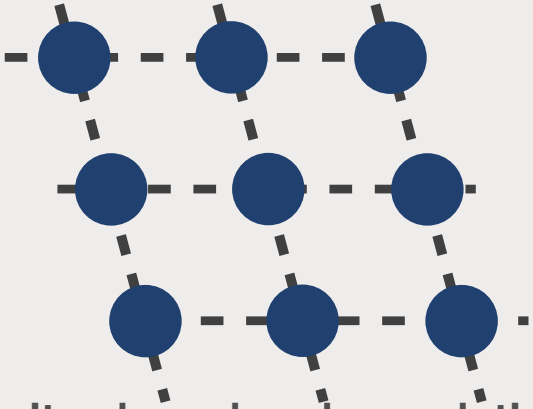
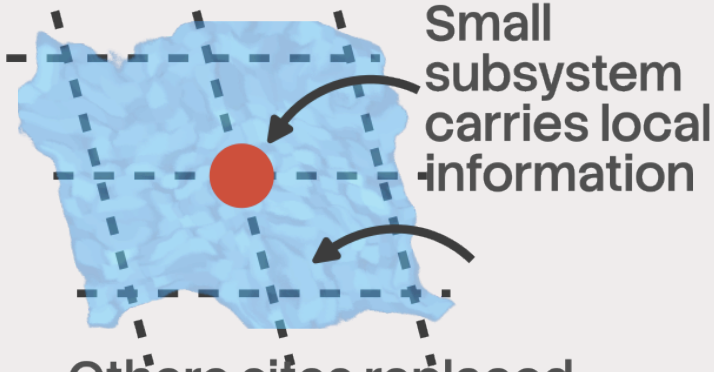


All sites have local correlation

**Idea**: replace full problem with a simpler **auxiliary model** that captures the local correlations.

Auxiliary model



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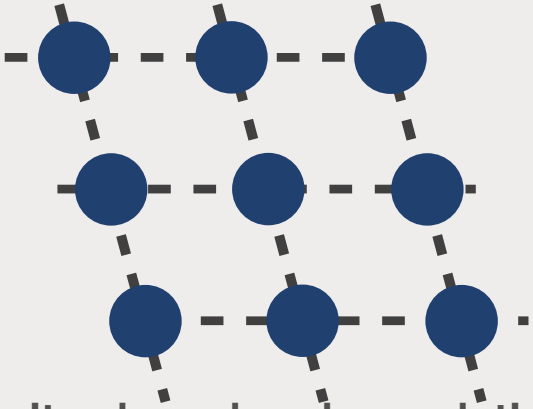

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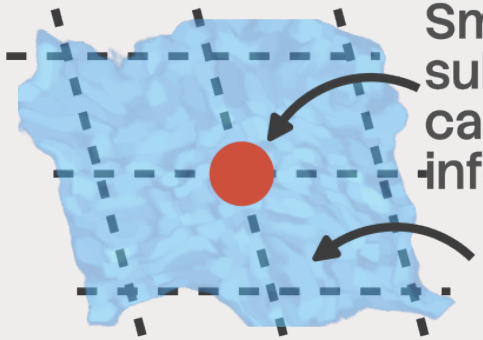

Small subsystem carries local information

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**Example: Mean-field solution of Ising model**

$$H = \sum_{i,j} J_{i,j} S_i^z S_j^z$$

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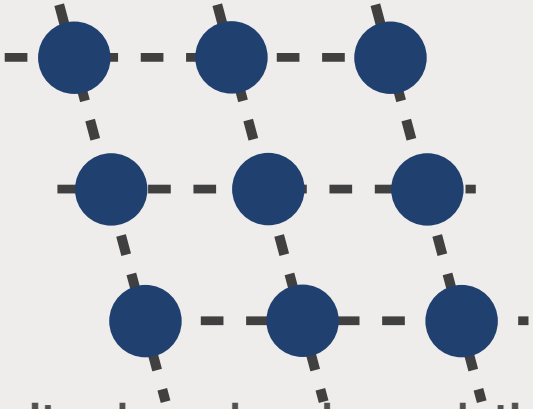

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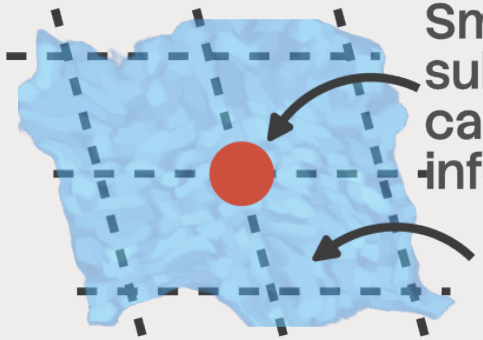

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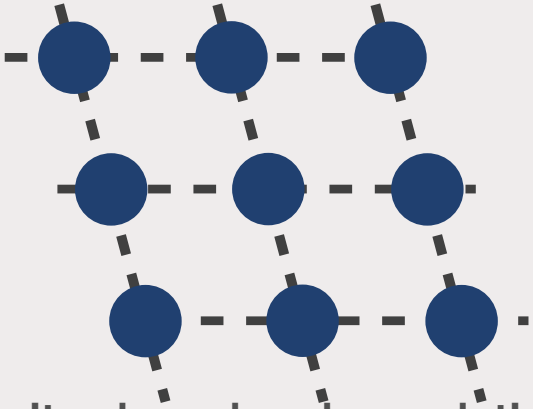

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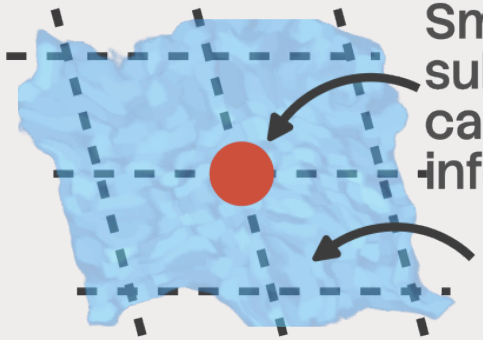

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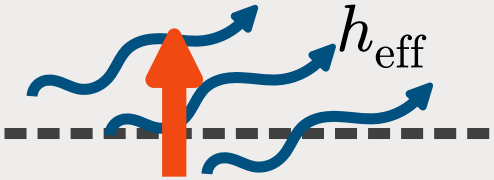



Environment replaced by effective field

# Quantum Impurity Models as Auxiliary Models

More details

- discuss Kondo screening
- discuss Kondo breakdown
- argue how LFL and LM can be thought of as metallic vs insulating behaviour

# Questions

- Questions about solving models
- Questions about developing methods

# Minimal Model For Kondo Breakdown

## A Route to Mott Localisation in Infinite Dimensions

**New J. Phys. 25 113011 (2023)**
AM, N S Vidhyadhiraja, A Taraphder, S Lal

# Capturing Spatial Fluctuations and Lattice Geometry

## Impurity-Lattice Mapping Beyond Mean-Field

**New J. Phys. 28 013503 (2026)**
AM, S R. Hassan, A Mukherjee, N S. Vidhyadhiraja, A Taraphder, S Lal

# Kondo Breakdown On The Square Lattice

## Revisiting the Mott Transition in Two Dimensions

**New J. Phys. 28 013503 (2026)**
AM, S R. Hassan, A Mukherjee, N S. Vidhyadhiraja, A Taraphder, S Lal

# Competing tendencies In a Multi-Orbital System

## Kondo screening vs. Mott localisation in a Heavy-Fermion model

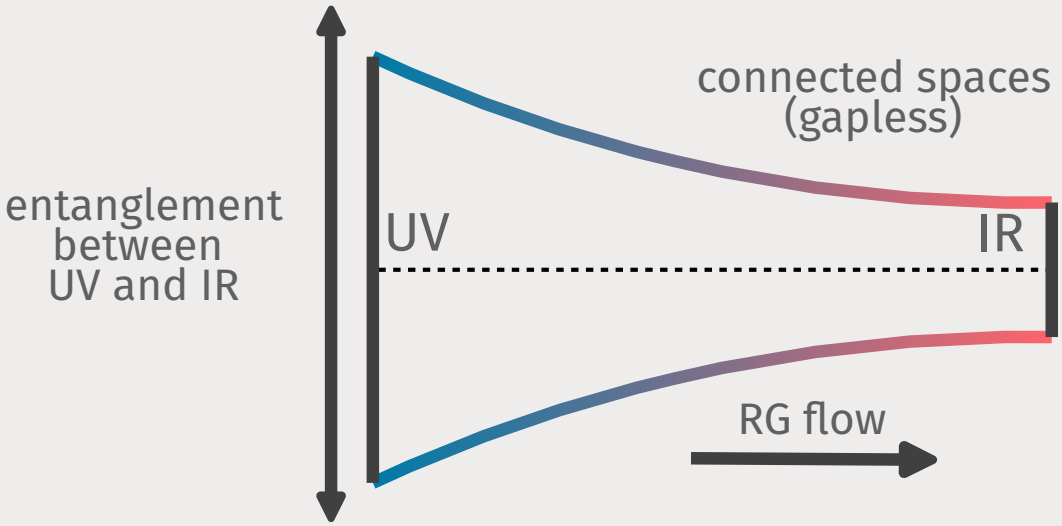
**(in progress)**

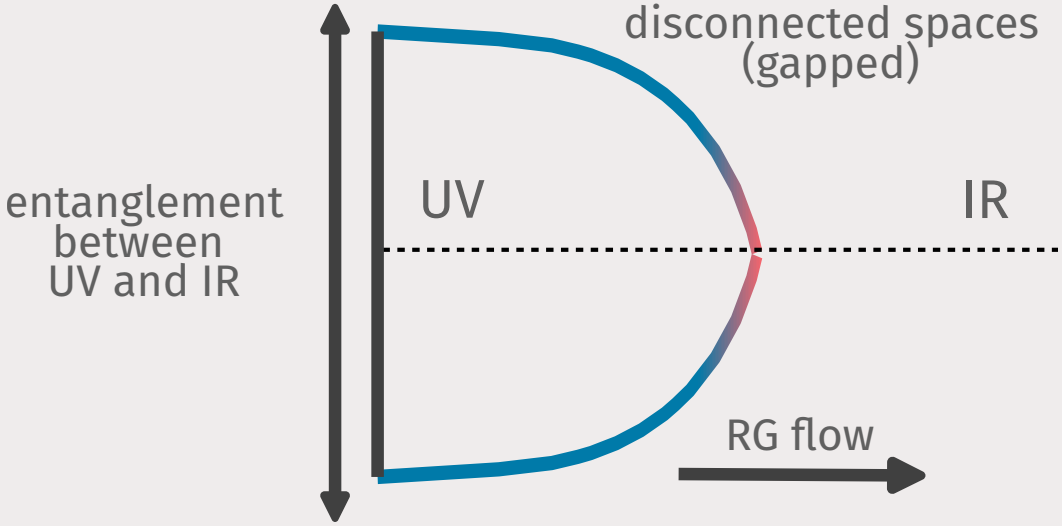
AM, S Lal

# Holography of Entanglement in 2D Free Fermions

## Emergent Geometry and Fermionic Criticality

**J. Phys. A: Math. Theor. 57 275401 (2024)**
AM, S Patra, S Lal


connected spaces
(gapless)
entanglement
between
UV and IR
UV
IR
RG flow


disconnected spaces
(gapped)
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# Closing Remarks

# Summary of Primary Results

# Main Conclusions and Takeaways

# Future Work